

| #    | 10.8 | 10.20 | 10.21 | 10.23 | 10.49 | 10.50 | 10.53 | 10.83 | 10.95 | 10.97 | 10.106 | 10.111 |
|------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------|--------|
| Page | 495  | 504   | 504   | 505   | 512   | 516   | 516   | 539   | 547   | 547   | 554    | 555    |

**\*10.8** A two-stage clinical trial is planned for testing  $H_0: p = .10$  versus  $H_a: p > .10$ , where  $p$  is the proportion of responders among patients who were treated by the protocol treatment. At the first stage, 15 patients are accrued and treated. If 4 or more responders are observed among the (first) 15 patients,  $H_0$  is rejected, the study is terminated, and no more patients are accrued. Otherwise, another 15 patients will be accrued and treated in the second stage. If a total of 6 or more responders are observed among the 30 patients accrued in the two stages (15 in the first stage and 15 more in the second stage), then  $H_0$  is rejected. For example, if 5 responders are found among the first-stage patients,  $H_0$  is rejected and the study is over. However, if 2 responders are found among the first-stage patients, 15 second-stage patients are accrued, and an additional 4 or more responders (for a total of 6 or more among the 30) are identified,  $H_0$  is rejected and the study is over.<sup>1</sup>

- Use the binomial table to find the numerical value of  $\alpha$  for this testing procedure.
- Use the binomial table to find the probability of rejecting the null hypothesis when using this rejection region if  $p = .30$ .
- For the rejection region defined above, find  $\beta$  if  $p = .30$ .

$X$  = no responders in Stage 1

$Y$  = " stage 2

a) Probability function that  $X \geq 4$  or  $X+Y \geq 6$  when  $p=0.1$

$$\Rightarrow P(X \geq 4) = 1 - P(X \leq 3) = 1 - 0.94 = 0.056$$

Numerical value: if  $X < 4$ , need to calculate  $X+Y \geq 6$

$$\begin{aligned} \Rightarrow P(X \leq 3, X+Y \geq 6) &= P(X=0)P(Y \geq 6) + P(X=1)P(Y \geq 5) + P(X=2)P(Y \geq 4) + P(X=3)P(Y \geq 3) \\ &= 0.2059 \times 0.002 + 0.3431 \times 0.013 + 0.2669 \times 0.056 + 0.1285 \times 0.164 \\ &= 0.004 + 0.0045 + 0.0149 + 0.0237 \\ &= 0.0435 \end{aligned}$$

$$\alpha = P(X \geq 4) + P(X \leq 3, X+Y \geq 6) = 0.056 + 0.0435 = 0.0995$$

b) Type I error  $\rightarrow$  reject null hypothesis  $H_0$ , when  $H_0$  is true  $p=0.30$   $n=15$

- reject when 4+ responders

$$P(X \geq 4) = 1 - P(X \leq 3) = 1 - 0.797 = 0.703$$

Calculate probability: (move to stage 2)

$$\begin{aligned} P(X+Y \geq 6) &= P(X=0)P(Y \geq 6) + P(X=1)P(Y \geq 5) + P(X=2)P(Y \geq 4) + P(X=3)P(Y \geq 3) \\ &= 0.005(0.278) + 0.030(0.485) + 0.092(0.703) + 0.170(0.873) \\ &= 0.00139 + 0.01455 + 0.064676 + 0.14841 \\ &= 0.229026 \end{aligned}$$

$$\text{Probability of rejection } \alpha = 1 - 0.229026 = 0.770974$$

c) find  $\beta$  if  $p=0.30$

$$\begin{aligned} \beta &= P(X+Y \leq 5) = P(X=0)P(Y \leq 5) + P(X=1)P(Y \leq 4) + P(X=2)P(Y \leq 3) + P(X=3)P(Y \leq 2) \\ &= 0.005(0.722) + 0.030(0.515) + 0.092(0.297) + 0.170(0.127) \\ &= 0.00361 + 0.01545 + 0.027324 + 0.170(0.127) \\ &= 0.00361 + 0.01545 + 0.027324 + 0.02159 \\ &= 0.067974 \end{aligned}$$

- 10.20** The Rockwell hardness index for steel is determined by pressing a diamond point into the steel and measuring the depth of penetration. For 50 specimens of an alloy of steel, the Rockwell hardness index averaged 62 with standard deviation 8. The manufacturer claims that this alloy has an average hardness index of at least 64. Is there sufficient evidence to refute the manufacturer's claim at the 1% significance level?

← relatively large population      ← confirms

$$n = 50, \bar{X} = 62, s = 8, \mu_0 = 64, \alpha = 0.01$$

Alternative hypothesis =  $H_a: \mu < 64$  (only left-tailed)

\*Using t-test:

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} = \frac{62 - 64}{8/\sqrt{50}} = \frac{-2}{1.1314} \approx -1.7688$$

Deg of freedom =  $n - 1 = 49 \rightarrow$  left-tailed test at  $\alpha = 0.01 \rightarrow t_{0.01, 49} \approx -2.407$

$\Rightarrow -1.7688 > -2.407$ , test stat does not fall in the rejection region

- 10.21** Shear strength measurements derived from unconfined compression tests for two types of soils gave the results shown in the following table (measurements in tons per square foot). Do the soils appear to differ with respect to average shear strength, at the 1% significance level?

| Soil Type I        | Soil Type II       |
|--------------------|--------------------|
| $n_1 = 30$         | $n_2 = 35$         |
| $\bar{y}_1 = 1.65$ | $\bar{y}_2 = 1.43$ |
| $s_1 = 0.26$       | $s_2 = 0.22$       |

← large sample

2. Source: Kurt Grote, T. L. Lincoln, and J. G. Gamble, "Hip Adductor Injury in Competitive Swimmers," *American Journal of Sports Medicine* 32(1) (2004): 104.

two-sample z-test

$$z = \frac{\bar{y}_1 - \bar{y}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{1.65 - 1.43}{\sqrt{\frac{0.26^2}{30} + \frac{0.22^2}{35}}} = \frac{0.22}{\sqrt{0.002255 + 0.00138}} = \frac{0.22}{0.0603} = 3.648$$

since two-tailed  $\Rightarrow \alpha/2 = 0.005$

$$Z_{0.005} = \pm 2.576 \text{ (from table)}$$

Since  $|3.648| > 2.576$ , we reject  $H_0$  in favor of  $H_a$

$\Rightarrow$  Mean shear strengths of the two soil types are different

- 10.23** Studies of the habits of white-tailed deer indicate that these deer live and feed within very limited ranges, approximately 150 to 205 acres. To determine whether the ranges of deer located in two different geographical areas differ, researchers caught, tagged, and fitted 40 deer with small radio transmitters. Several months later, the deer were tracked and identified, and the distance  $y$  from the release point was recorded. The mean and standard deviation of the distances from the release point were as given in the accompanying table.<sup>3</sup>

|                                | Location |         |
|--------------------------------|----------|---------|
|                                | 1        | 2       |
| Sample size                    | 40       | 40      |
| Sample mean (ft)               | 2980     | 3205    |
| Sample standard deviation (ft) | 1140     | 963     |
| Population mean                | $\mu_1$  | $\mu_2$ |

- If you have no preconceived reason for believing that one population mean is larger than the other, what would you choose for your alternative hypothesis? Your null hypothesis?
- Would your alternative hypothesis in part (a) imply a one-tailed or a two-tailed test? Explain.
- Do the data provide sufficient evidence to indicate that the mean distances differ for the two geographical locations? Test using  $\alpha = .10$ .

- a) We have no reason to believe one location has greater distance than the other

Null hypothesis:  $H_0: \mu_1 = \mu_2$

Alternative hypothesis:  $H_a: \mu_1 \neq \mu_2$

- b) Since we are testing for any difference (not direction specific)  $\Rightarrow$  two-tailed test

- c) two-sample z-test

$$Z = \frac{2980 - 3705}{\sqrt{\frac{1140^2}{40} + \frac{963^2}{40}}} = \frac{-725}{\sqrt{32490 + 23184.75}} \approx -0.953$$

Since two-tailed  $\Rightarrow \frac{\alpha}{2} = 0.05$   
 $Z_{0.05} = \pm 1.645$

$|-0.953| < 1.645 \Rightarrow$  fail to reject the null hypothesis

$\Rightarrow$  There's not enough evidence at the 10% significance level to conclude that the avg distances differ between two locations

**10.49** Refer to Exercise 10.19. Construct a 95% upper confidence bound for the average voltage reading.

- a How does the value  $\mu = 130$  compare to this upper bound?  
 b Based on the upper bound in part (a), should the alternative hypothesis of Exercise 10.19 be accepted?  
 c Is there any conflict between the answer in part (b) and your answer to Exercise 10.19?

- a)  $n=40$ ,  $\bar{X} = 128.6$ ,  $s = 2.1$ ,  $\alpha = 0.05$  (for a one-sided 95% upper bound)

$$\text{Upper bound} = \bar{X} + t_{\alpha} \cdot \frac{s}{\sqrt{n}} = 128.6 + 1.685 \cdot \frac{2.1}{\sqrt{40}} \approx 129.16$$

$$130 > 129.16$$

$\Rightarrow \mu = 130$  is above the upper bound

- b)  $H_0: \mu = 130$  vs  $H_a: \mu < 130$

$\Rightarrow$  Since the entire confidence interval is below 130, this supports  $H_a$   
 $\Rightarrow$  alternative hypothesis should be accepted

- c) No, there's no conflict. Both the hypothesis test and the 95% upper confidence bound lead to rejecting the null hypothesis at the 5% significance level. Since 130 is above the upper bound, both methods support the conclusion that the true mean is less than 130.

**10.50** High airline occupancy rates on scheduled flights are essential for profitability. Suppose that a scheduled flight must average at least 60% occupancy to be profitable and that an examination of the occupancy rates for 120 10:00 A.M. flights from Atlanta to Dallas showed mean occupancy rate per flight of 58% and standard deviation 11%. Test to see if sufficient evidence exists to support a claim that the flight is unprofitable. Find the  $p$ -value associated with the test. What would you conclude if you wished to implement the test at the  $\alpha = .10$  level?

$$p_0 = 0.60, \bar{x} = 0.58, s = 0.11, n = 120, \alpha = 0.10$$

Null hyp:  $H_0: \mu = 0.60$   
 Alt hyp:  $H_a: \mu < 0.60$

$$Z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{0.58 - 0.60}{0.11/\sqrt{120}} \approx -1.99 \Rightarrow p\text{-value} = P(Z < -1.99) \approx 0.0232 < 0.10 = \alpha$$

$\Rightarrow$  Reject  $H_0$  at 10% level of significance

How would you like to live to be 200 years old? For centuries, humankind has sought the key to the mystery of aging. What causes aging? How can aging be slowed? Studies have focused on *biomarkers*, physical or biological changes that occur at a predictable time in a person's life. The theory is that, if ways can be found to delay the occurrence of these biomarkers, human life can be extended. A key biomarker, according to scientists, is forced vital capacity (FVC), the volume of air that a person can expel after taking a deep breath. A study of 5209 men and women aged 30 to 62 showed that FVC declined, on the average, 3.8 deciliters (dl) per decade for men and 3.1 deciliters per decade for women.<sup>10</sup> Suppose that you wished to determine whether a physical fitness program for men and women aged 50 to 60 would delay aging; to do so, you measured the FVC for 30 men and 30 women participating in the fitness program at the beginning and end of the 50- to 60-year age interval and recorded the drop in FVC for each person. A summary of the data appears in the accompanying table.

|                                 | Men     | Women   |
|---------------------------------|---------|---------|
| Sample size                     | 30      | 30      |
| Sample average drop in FVC (dl) | 3.6     | 2.7     |
| Sample standard deviation (dl)  | 1.1     | 1.2     |
| Population mean drop in FVC     | $\mu_1$ | $\mu_2$ |

10. Source: T. Boddé, "Biomarkers of Aging: Key to a Younger Life," *Bioscience* 31(8) (1981): 566-567.

- Do the data provide sufficient evidence to indicate that the decrease in the mean FVC over the decade for the men on the physical fitness program is less than 3.8 dl? Find the attained significance level for the test.
- Refer to part (a). If you choose  $\alpha = .05$ , do the data support the contention that the mean decrease in FVC is less than 3.8 dl?
- Test to determine whether the FVC drop for women on the physical fitness program was less than 3.1 dl for the decade. Find the attained significance level for the test.
- Refer to part (c). If you choose  $\alpha = .05$ , do the data support the contention that the mean decrease in FVC is less than 3.1 dl?

a)  $H_0: \mu = 3.8$        $n = 30, \bar{X} = 3.6, s = 1.1$   
 $H_a: \mu < 3.8$

$$Z = \frac{\sqrt{n}(\bar{X} - \mu)}{s} = \frac{\sqrt{30}(3.6 - 3.8)}{1.1} = -0.99$$

$$p\text{-value} = P(Z \leq -0.99) = 0.164$$

⇒ The attained level of significance is 16.4% with  $p\text{-value} = 0.164$

b)  $0.164 > 0.05 \Rightarrow$  fail to reject  $H_0$

⇒ The data doesn't support the contention that the mean FVC drop for men is less than 3.8 dl

c)  $H_0: \mu = 3.1$        $n = 30, \bar{X} = 2.7, s = 1.2$   
 $H_a: \mu < 3.1$

$$Z = \frac{\sqrt{n}(\bar{X} - \mu)}{s} = \frac{\sqrt{30}(2.7 - 3.1)}{1.2} = -1.826$$

$$\Rightarrow p\text{-value} = 0.0336$$

⇒ attained level of significance is 3.36%

d)  $0.0336 < 0.05 \Rightarrow$  reject  $H_0$

⇒ The data supports the claim that the mean FVC drop for women is less than 3.1 dl

## 10.83

The manager of a dairy is in the market for a new bottle-filling machine and is considering machines manufactured by companies A and B. If ruggedness, cost, and convenience are comparable in the two machines, the deciding factor will be the variability of fills (the machine producing fills with the smaller variance being preferable). Let  $\sigma_1^2$  and  $\sigma_2^2$  be the fill variances for machines produced by companies A and B, respectively. Now consider various tests of the null hypothesis  $H_0: \sigma_1^2 = \sigma_2^2$ . Obtaining samples of fills from the two machines and using the test statistic  $S_1^2/S_2^2$ , we could set up as the rejection region an upper-tail area, a lower-tail area, or a two-tailed area of the  $F$  distribution, depending on the interests to be served. Identify the type of rejection region that would be most favored by the following persons, and explain why.

- The manager of the dairy
- A salesperson for company A
- A salesperson for company B

a) Manager does not care which company has lower variability, only wants to know if there is any difference in variability  
 ⇒ two-tailed  $F$ -test

b) Company A salesperson wants to convince Company A's machine is better (smaller variance)  $\Rightarrow \sigma_1^2 < \sigma_2^2$   
 ⇒ for  $F = S_1^2/S_2^2$ , a small value supports  $\sigma_1^2 < \sigma_2^2$  ⇒ Lower-tail area

c) " B " " "  $\Rightarrow \sigma_2^2 < \sigma_1^2$   
 ⇒ for  $F = S_1^2/S_2^2$ , a bigger value supports  $\sigma_1^2 < \sigma_2^2$  ⇒ Upper-tail area



**10.95** Suppose that we have a random sample of four observations from the density function

$$f(y|\theta) = \begin{cases} \left(\frac{1}{2\theta^3}\right) y^2 e^{-y/\theta}, & y > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

- a Find the rejection region for the most powerful test of  $H_0: \theta = \theta_0$  against  $H_a: \theta = \theta_a$ , assuming that  $\theta_a > \theta_0$ . [Hint: Make use of the  $\chi^2$  distribution.]  
b Is the test given in part (a) uniformly most powerful for the alternative  $\theta > \theta_0$ ?

a) for  $Y_1, Y_2, Y_3, Y_4$ , we have Gamma distribution with shape  $\alpha=3$  and scale  $\theta$

\* Neyman-Pearson Lemma:  $\frac{f_0}{f_a} < k$

$$\frac{f_0}{f_a} = \frac{L(\theta_0)}{L(\theta_a)} = \frac{\left(\frac{1}{2\theta_0^3}\right) \exp\left(-\frac{1}{\theta_0} \sum_{i=1}^4 Y_i\right) \prod_{i=1}^4 Y_i^2}{\left(\frac{1}{2\theta_a^3}\right) \exp\left(-\frac{1}{\theta_a} \sum_{i=1}^4 Y_i\right) \prod_{i=1}^4 Y_i^2} = \left(\frac{\theta_a}{\theta_0}\right)^3 \exp\left(\frac{1}{\theta_a} \sum_{i=1}^4 Y_i - \frac{1}{\theta_0} \sum_{i=1}^4 Y_i\right)$$

$\theta_a$  and  $\theta_0$  are known constants so rejection region:

$$\exp\left(\frac{1}{\theta_a} \sum_{i=1}^4 Y_i - \frac{1}{\theta_0} \sum_{i=1}^4 Y_i\right) < k$$

$$\frac{\theta_0 - \theta_a}{\theta_0 \theta_a} \sum_{i=1}^4 Y_i < \ln(k)$$

$$\sum_{i=1}^4 Y_i > \frac{\theta_0 - \theta_a}{\theta_0 \theta_a} \ln(k) \rightarrow \text{all constants}$$

$$\sum_{i=1}^4 Y_i > c \leftarrow \text{constant that can be found with significance level test}$$

$$\text{Each } Y_i \sim \text{Gamma}(3, \theta) \rightarrow \sum_{i=1}^4 Y_i \sim \text{Gamma}(12, \theta)$$

$$\Rightarrow \frac{2}{\theta} \sum_{i=1}^4 Y_i \sim \chi_{24}^2$$

$$P\left(\sum_{i=1}^4 Y_i > c\right) = \alpha$$

$$P\left(\sum_{i=1}^4 Y_i \frac{2}{\theta_0} > \frac{2c}{\theta_0}\right) = \alpha$$

$$\frac{2c}{\theta_0} = \chi_{24, \alpha}^2$$

$$c = \frac{\theta_0}{2} \chi_{24, \alpha}^2$$

$$\Rightarrow \text{Reject } H_0 \text{ if } \sum_{i=1}^4 Y_i > \frac{\theta_0}{2} \chi_{24, \alpha}^2$$

b) The rejection region found in a) does not depend on  $\theta^2$   
 $\Rightarrow$  this is a UMP test for testing  $H_0: \theta = \theta_0$  vs  $H_a: \theta > \theta_0$

\* Also, gamma distribution has MLR (monotone likelihood ratio) property in statistic  $\sum Y_i$  when testing  $\theta > \theta_0$ .  
 $\Rightarrow$  Probability of large values of  $\sum Y_i$  increases as  $\theta$  increases

**\*10.97** Let  $Y_1, Y_2, \dots, Y_n$  be independent and identically distributed random variables with discrete probability function given by

|               | y          |                     |                |
|---------------|------------|---------------------|----------------|
|               | 1          | 2                   | 3              |
| $p(y \theta)$ | $\theta^2$ | $2\theta(1-\theta)$ | $(1-\theta)^2$ |

where  $0 < \theta < 1$ . Let  $N_i$  denote the number of observations equal to  $i$  for  $i = 1, 2, 3$ .

- a Derive the likelihood function  $L(\theta)$  as a function of  $N_1, N_2$ , and  $N_3$ .  
b Find the most powerful test for testing  $H_0: \theta = \theta_0$  versus  $H_a: \theta = \theta_a$ , where  $\theta_a > \theta_0$ . Show that your test specifies that  $H_0$  be rejected for certain values of  $2N_1 + N_2$ .  
c How do you determine the value of  $k$  so that the test has nominal level  $\alpha$ ? You need not do the actual computation. A clear description of how to determine  $k$  is adequate.  
d Is the test derived in parts (a)-(c) uniformly most powerful for testing  $H_0: \theta = \theta_0$  versus  $H_a: \theta > \theta_0$ ? Why or why not?

$$a) L(\theta) = \prod_{i=1}^n p(Y_i|\theta) = \theta^{2N_1} \cdot (2\theta(1-\theta))^{N_2} \cdot (1-\theta)^{2N_3} = 2^{N_2} \theta^{2N_1+N_2} (1-\theta)^{2N_3+N_2}$$

$$b) \frac{f_0}{f_a} = \frac{L(\theta_0)}{L(\theta_a)} = \frac{2^{N_2} \theta_0^{2N_1+N_2} (1-\theta_0)^{2N_3+N_2}}{2^{N_2} \theta_a^{2N_1+N_2} (1-\theta_a)^{2N_3+N_2}} = \left(\frac{\theta_0}{\theta_a}\right)^{2N_1+N_2} \left(\frac{1-\theta_0}{1-\theta_a}\right)^{N_2+2N_3}$$

\* This ratio is monotonic in  $T = 2N_1 + N_2$  because as  $T$  increases,  $\theta_0/\theta_a < 1$  leads to ratio decreasing  
 $\Rightarrow$  rejection region is reject  $H_0$  if  $2N_1 + N_2 > c$  where  $c$  is a constant chosen to achieve level  $\alpha$

c) To ensure the test has nominal level  $\alpha$ , choose the smallest value of  $k$  for which probability of type I error is  $\leq \alpha$

$\rightarrow$  smallest  $k$  such that  $P(Z_{N_1+N_2} > k) \leq \alpha$

d) Yes, it is UMP because it doesn't depend on  $\theta$  and the likelihood ratio is monotone in  $T = Z_{N_1+N_2}$

**10.106** A survey of voter sentiment was conducted in four midcity political wards to compare the fraction of voters favoring candidate A. Random samples of 200 voters were polled in each of the four wards, with the results as shown in the accompanying table. The numbers of voters favoring A in the four samples can be regarded as four independent binomial random variables.

Construct a likelihood ratio test of the hypothesis that the fractions of voters favoring candidate A are the same in all four wards. Use  $\alpha = .05$ .

| Opinion        | Ward |     |     |     | Total |
|----------------|------|-----|-----|-----|-------|
|                | 1    | 2   | 3   | 4   |       |
| Favor A        | 76   | 53  | 59  | 48  | 236   |
| Do not favor A | 124  | 147 | 141 | 152 | 564   |
| Total          | 200  | 200 | 200 | 200 | 800   |

$H_0: p_1 = p_2 = p_3 = p_4 = p$

$H_a: \text{At least 1 } p_i \text{ differs}$

Find value of  $L(S)$  and  $L(S_0)$

$$\hat{p}_i = \frac{y_i}{200} \leftarrow \text{MLE of } p_i$$

$$L(S) = \prod_{i=1}^4 \binom{200}{y_i} p_i^{y_i} (1-p_i)^{200-y_i} = \prod_{i=1}^4 \binom{200}{y_i} \left(\frac{y_i}{200}\right)^{y_i} \left(1 - \frac{y_i}{200}\right)^{200-y_i}$$

$$L(S_0) = \prod_{i=1}^4 \binom{200}{y_i} p^{y_i} (1-p)^{200-y_i} = \prod_{i=1}^4 \binom{200}{y_i} p^{\sum y_i} (1-p)^{800 - \sum y_i}$$

$\rightarrow$  As  $p = p$   $y_i = 76, 53, 59, 48$

Take log likelihood:  $l(p) = \sum_{i=1}^4 y_i \log(p) + (800 - \sum_{i=1}^4 y_i) \log(1-p) + c$

Find optimum:  $\frac{dl}{dp} = \frac{\sum_{i=1}^4 y_i}{p} - \frac{800 - \sum_{i=1}^4 y_i}{1-p} = 0$

$$800 p - \sum_{i=1}^4 y_i = \sum_{i=1}^4 y_i - p \sum_{i=1}^4 y_i$$

$$\hat{p} = \frac{\sum_{i=1}^4 y_i}{800}$$

$$L(S_0) = \prod_{i=1}^4 \binom{200}{y_i} \left(\frac{\sum_{i=1}^4 y_i}{800}\right)^{y_i} \left(1 - \frac{\sum_{i=1}^4 y_i}{800}\right)^{200-y_i}$$

$$\text{Likelihood ratio: } \lambda = \frac{L(S_0)}{L(S)} = \frac{\prod_{i=1}^4 \binom{200}{y_i} \left(\frac{\sum_{i=1}^4 y_i}{800}\right)^{y_i} \left(1 - \frac{\sum_{i=1}^4 y_i}{800}\right)^{200-y_i}}{\prod_{i=1}^4 \binom{200}{y_i} \left(\frac{y_i}{200}\right)^{y_i} \left(1 - \frac{y_i}{200}\right)^{200-y_i}}$$

\*  $S_0 \rightarrow 3$  parameters and  $S \rightarrow 0 \rightarrow DF = 3$

$$\begin{aligned} \log(\lambda) &= \sum_{i=1}^4 y_i \cdot \log\left(\frac{\sum y_i}{800}\right) + (800 - \sum y_i) \cdot \log\left(1 - \frac{\sum y_i}{800}\right) - \sum_{i=1}^4 y_i \cdot \log\left(\frac{y_i}{200}\right) - \sum_{i=1}^4 (200 - y_i) \cdot \log\left(1 - \frac{y_i}{200}\right) \\ &= 236 \times 4.08 + 564 \times 4.95 - 329.136 - 210.425 - 240.575 - 185.817 - 597.715 - 733.594 - 697.775 - 763.63 \\ &= -3.987 \end{aligned}$$

test statistic is  $-2 \log(\lambda) \approx \chi^2_{(k-1)}$  where  $k=4$

$$-2 \log(\lambda) = -2 \log(-3.987) = 7.974$$

$$\chi^2_{0.05, 3} = 7.815 < 7.974$$

$\rightarrow$  We reject  $H_0$  at 5% level

**10.111** Suppose that we are interested in testing the simple null hypothesis  $H_0: \theta = \theta_0$  versus the simple alternative hypothesis  $H_a: \theta = \theta_a$ . According to the Neyman-Pearson lemma, the test that maximizes the power at  $\theta_a$  has a rejection region determined by

$$\frac{L(\theta_0)}{L(\theta_a)} < k.$$

In the context of a likelihood ratio test, if we are interested in the simple  $H_0$  and  $H_a$ , as stated, then  $\Omega_0 = \{\theta_0\}$ ,  $\Omega_a = \{\theta_a\}$ , and  $\Omega = \{\theta_0, \theta_a\}$ .

a Show that the likelihood ratio  $\lambda$  is given by

$$\lambda = \frac{L(\theta_0)}{\max\{L(\theta_0), L(\theta_a)\}} = \frac{1}{\max\left\{1, \frac{L(\theta_a)}{L(\theta_0)}\right\}}.$$

b Argue that  $\lambda < k$  if and only if, for some constant  $k'$ ,

$$\frac{L(\theta_0)}{L(\theta_a)} < k'.$$

c What do the results in parts (a) and (b) imply about likelihood ratio tests when both the null and alternative hypotheses are simple?

$$a) \lambda = \frac{L(\theta_0)}{L(\theta_1)} = \frac{L(\theta_0)}{\max\{L(\theta_0), L(\theta_1)\}} = \frac{1}{\max\left\{\frac{L(\theta_0)}{L(\theta_1)}, \frac{L(\theta_1)}{L(\theta_0)}\right\}} = \frac{1}{\max\left\{1, \frac{L(\theta_1)}{L(\theta_0)}\right\}}$$

b) from part a), we know that:

$$\lambda = \begin{cases} 1 & \text{if } L(\theta_0) \geq L(\theta_1) \\ \frac{L(\theta_1)}{L(\theta_0)} & \text{if } L(\theta_0) < L(\theta_1) \end{cases}$$

$$\text{If } \lambda < k, \text{ and } L(\theta_1) > L(\theta_0), \text{ then: } \lambda = \frac{L(\theta_0)}{L(\theta_1)} < k \Rightarrow \frac{L(\theta_0)}{L(\theta_1)} < k' \text{ where } k' = k$$

$\Rightarrow$  Condition  $\lambda < k$  implies likelihood ratio  $\frac{L(\theta_0)}{L(\theta_1)}$  is less than some constant  $k'$ , only in region where  $H_0$  is more likely than  $H_1$

c) From a) and b), we realize that this directly matches the form of the Neyman-Pearson Lemma: For testing a simple null hyp.  $H_0: \theta = \theta_0$  against a simple alternative  $H_1: \theta = \theta_1$ , the most powerful test at level  $\alpha$  is the one that rejects  $H_0$  when the likelihood  $\frac{L(\theta_0)}{L(\theta_1)}$  is small enough

$\Rightarrow$  Likelihood ratio test is the most powerful test for distinguishing between two simple hypotheses